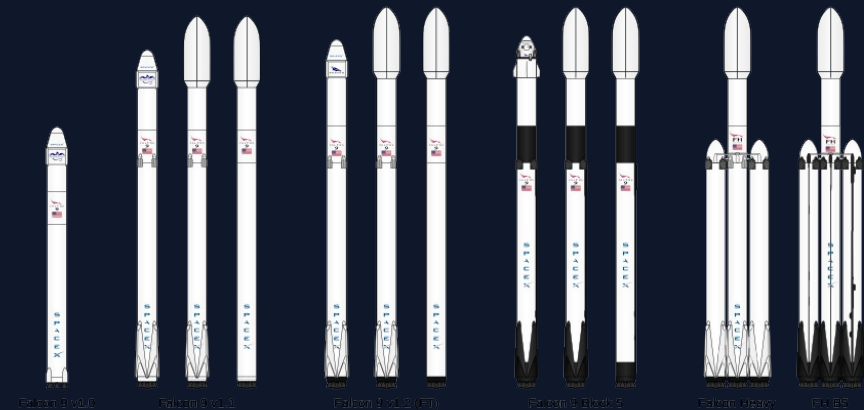


SpaceX Falcon 9 Launch Success Prediction

Applied Data Science Capstone final report

Finnred Fernandes

An end-to-end analysis covering data collection, wrangling, SQL, interactive visual analytics, dashboards, and machine learning.



Key outputs: 90 launches analyzed • 67% success rate • 80 engineered features

Executive summary

The capstone combined historical SpaceX launch data, visual analytics, and classification models to understand and predict first-stage landing success.

- The working dataset contains 90 Falcon 9 launches, 3 launch sites, and 11 orbit classes after cleaning and target-label creation.
- Landing success improved over time and was strongest at KSC LC-39A and VAFB SLC 4E, while GTO missions remained the most challenging.
- Interactive analytics (Folium + Dash) made the findings easy to explore, while machine learning turned the insights into a predictive workflow.

90

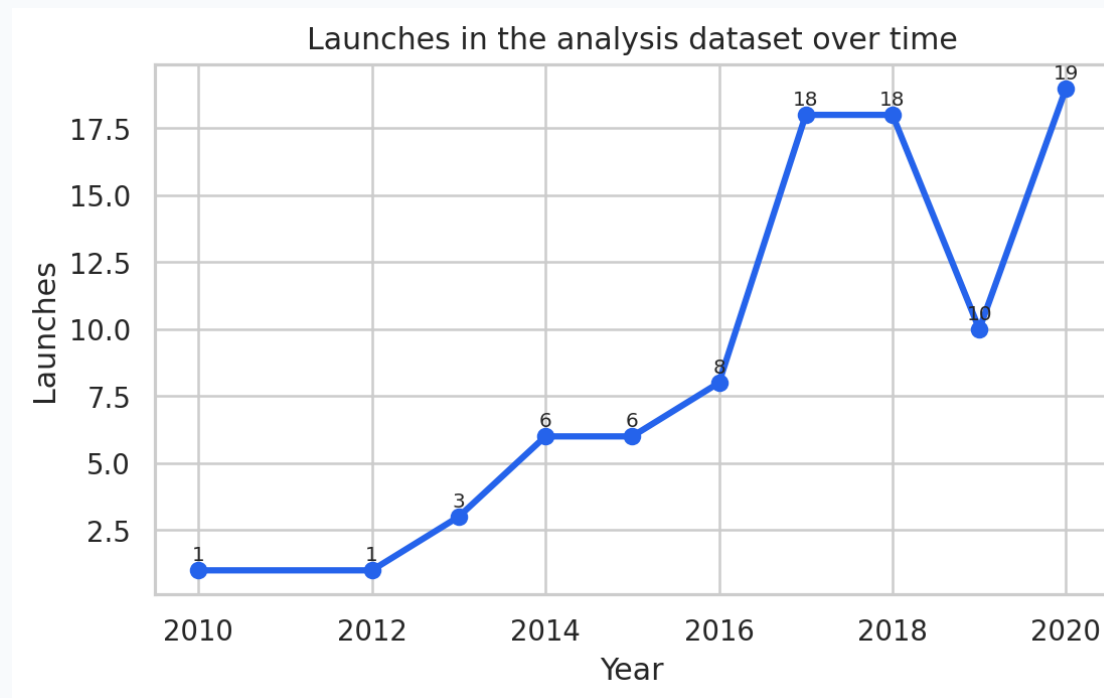
Falcon 9 launches analyzed

67%

overall landing success

80

one-hot encoded features



The number of launches rose sharply in 2017-2020, creating a richer learning signal for both descriptive and predictive analysis.

Business problem and analytical workflow

SpaceX advertises reusable launches at a lower price point, so accurately understanding landing success matters for cost, cadence, and mission planning.

Objective

Predict whether the Falcon 9 first stage lands successfully and explain the operational patterns that drive that outcome.

Why it matters

- A successful landing lowers the cost of reusing boosters and improves future launch economics.
- Operationally, launch site, orbit, and payload affect the probability of a controlled return.

Final report structure mirrors the IBM capstone rubric: methodology + results for collection, wrangling, EDA, SQL, map, dashboard, and predictive analysis.

1 Collect

SpaceX API, Wikipedia scraping, and course CSVs

2 Wrangle

Clean fields, handle nulls, build class labels

3 EDA

Charts and SQL

4 Interactive analytics

Folium mapping and Plotly Dash filters

5 Modeling

Feature engineering, scaling, train/test split, GridSearchCV

Decision support

Outcome
A technically detailed report that explains historical patterns, shows interactive outputs, and demonstrates a predictive model for landing success.

Data collection methodology

The project assembled launch information from multiple sources so the later visual and predictive steps all used the same SpaceX mission context.

1 SpaceX API

Historical launches plus related fields such as rocket, payload, core, and launchpad metadata.

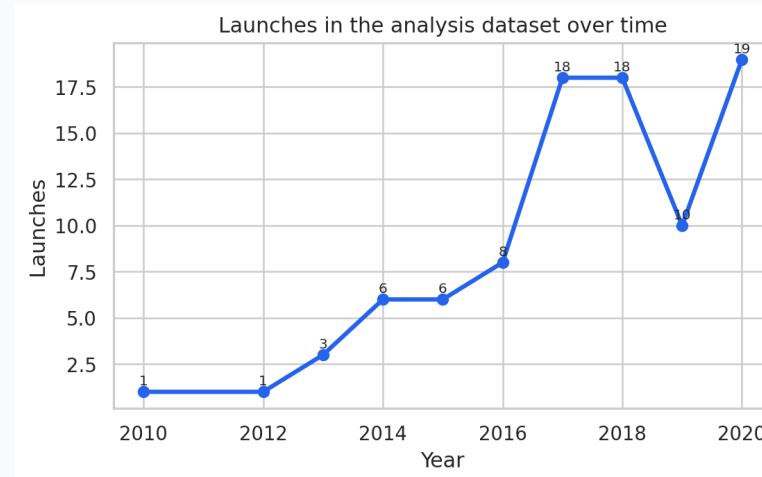
2 Wikipedia scraping

Falcon 9 / Falcon Heavy launch tables parsed with BeautifulSoup for additional launch-record context.

3 Course data assets

Curated CSVs used for EDA, Folium, Dash, SQL, and machine learning sections of the capstone.

Unified result: one Falcon 9 analysis dataset with 17 core fields before label creation and feature engineering.



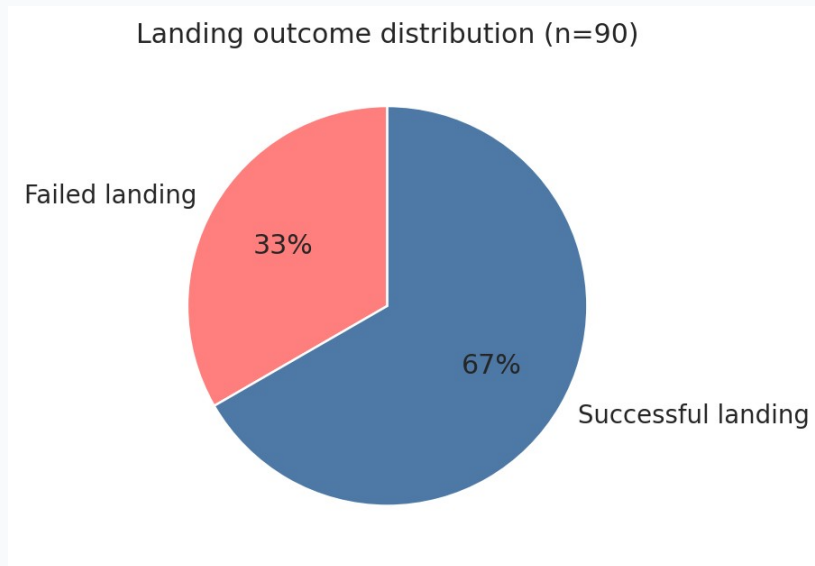
The launch count accelerates after 2016, which is why later years dominate the visual trend lines and model training signal.

Collection outputs used

Data asset	Rows / purpose
dataset_part_1.csv	90 launches, 17 columns, core launch records
Spacex.csv	101 rows, SQL practice and aggregation queries
spacex_launch_geo.csv	56 rows, map coordinates + class labels
spacex_launch_dash.csv	56 rows, interactive dashboard data

Data wrangling methodology

Wrangling turned raw launch outcomes into a supervised-learning target and prepared the data for consistent analysis across notebooks.



Main wrangling steps

- Created the binary target variable Class: successful outcomes mapped to 1 and failed / attempted outcomes mapped to 0.
- Verified launch-site and orbit distributions and preserved the numeric fields required later for visualization and machine learning.
- Retained 90 Falcon 9 records and used the course-defined outcome rules so quiz answers and later labs stay consistent.

Examples of successful outcomes

True ASDS
True RTLS
True Ocean

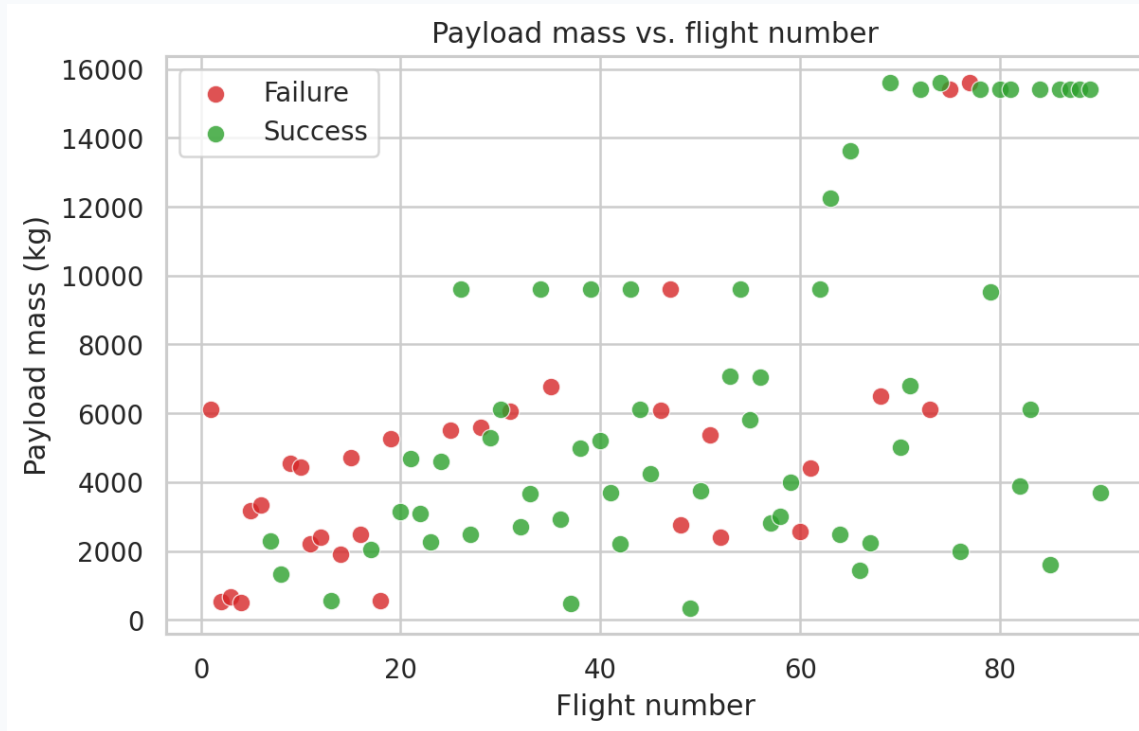
Mapped to the failure class

False Ocean
False ASDS
False RTLS
None ASDS / None None

Resulting class balance: 60 successful vs 30 failed launches, or roughly a 2:1 split.

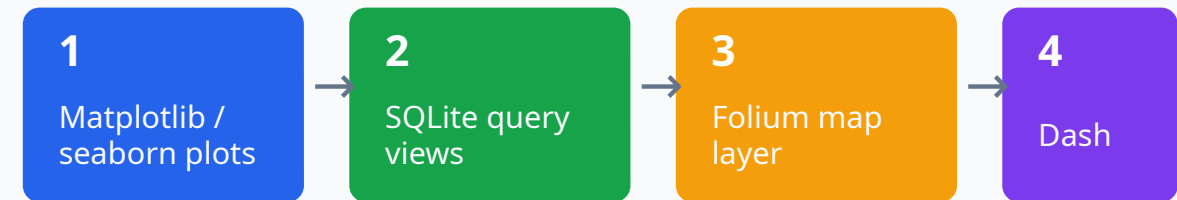
EDA and interactive visual analytics methodology

The analytical questions progressed from descriptive plots to SQL, geospatial analysis, and dashboard interactions.



Questions answered during EDA

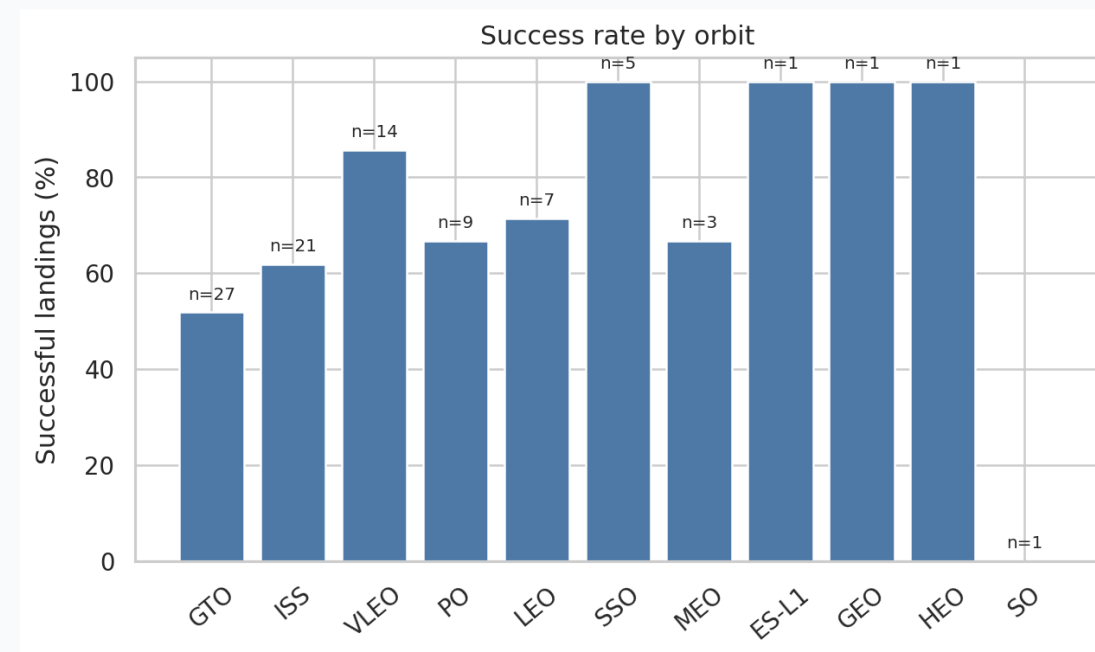
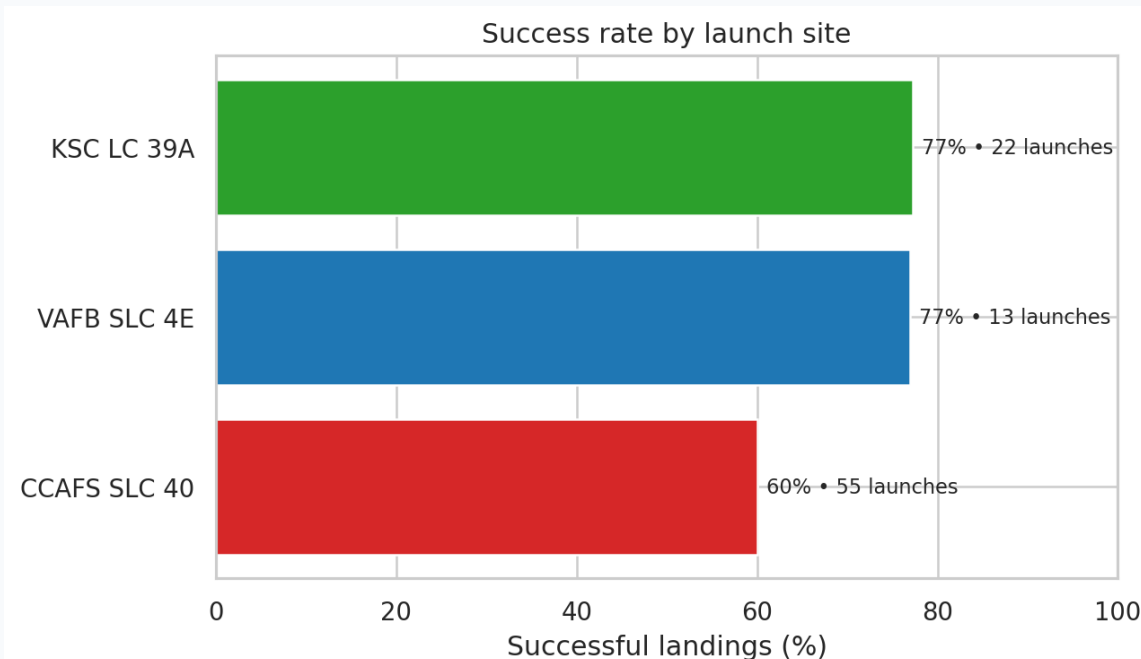
- How did landing performance change as SpaceX flew more missions and carried heavier payloads?
- Which launch sites and orbit classes were most associated with successful returns?
- What complementary views could SQL, Folium, and Dash add beyond static charts?



The scatter plot immediately shows why EDA mattered: payload alone does not determine success; trajectory and operational context matter too.

EDA with visualization results

Site and orbit views surfaced the main historical patterns before moving into SQL and interactive tools.



Key finding: KSC LC-39A and VAFB SLC 4E delivered about 77% success, while VLEO / SSO / GEO-class missions were more favorable than GTO missions.

EDA with SQL results

SQL provided a compact way to validate site counts, payload summaries, and landing milestones directly from tabular mission records.

Representative query pattern

```
SELECT Launch_Site,
COUNT(Mission_Outcome)
AS
mission_outcome_count
FROM SPACEXTBL
GROUP BY Launch_Site;
```

Why SQL added value

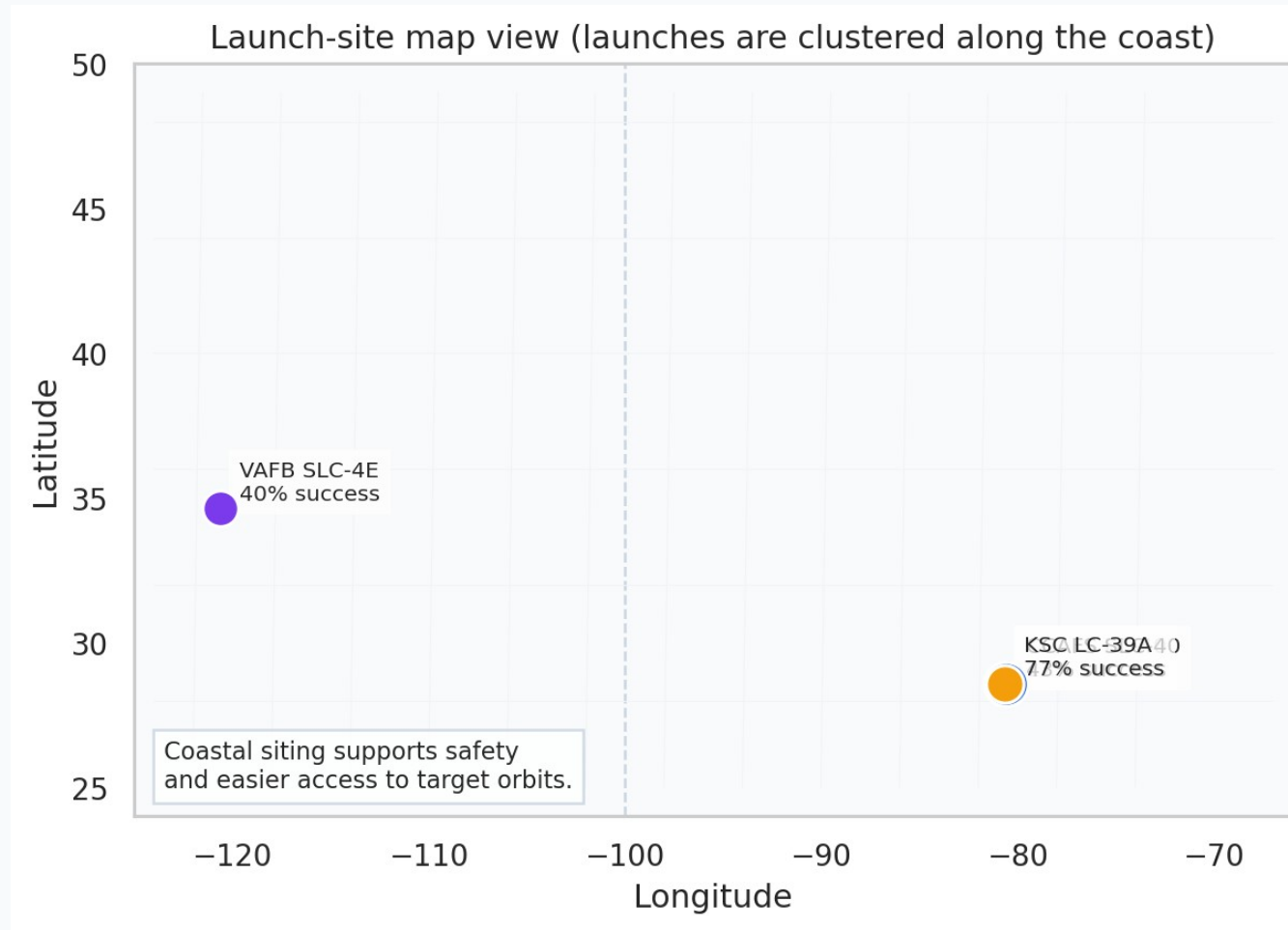
- Fast validation of payload and site-level summaries
- Easy grouping for mission outcomes and filtered subsets
- A reproducible bridge between raw records and later visuals

Query insight	Result
Distinct launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total NASA CRS payload	48,213 kg
Average payload (F9 v1.1)	2,928 kg
First successful ground-pad landing	2015-12-22
Mission outcomes by site	CCAFS LC-40: 26 • CCAFS SLC-40: 34 • KSC LC-39A: 16 • VAFB SLC-4E: 25

These outputs aligned the SQL notebook with the same story seen in the plots: payload profile, launch site, and landing context all matter.

Interactive map results using Folium

Geospatial analysis showed that SpaceX launch sites are positioned along the coast, balancing safety requirements with orbital access.



Location-based observations

- All analyzed launch sites are coastal, which reduces risk over populated areas and supports targeted orbital paths.
- KSC LC-39A and CCAFS SLC 40 cluster in Florida for eastward launches; VAFB SLC 4E supports polar / sun-synchronous trajectories from California.
- Mapped markers make site

Site snapshot

CCAFS SLC 40: 55 launches, 60% success
KSC LC 39A: 22 launches, 77% success
VAFB SLC 4E: 13 launches, 77% success

Plotly Dash dashboard results

The dashboard turned the same launch data into an exploratory interface with filters for launch site and payload range.

SpaceX Launch Records Dashboard

Launch Site

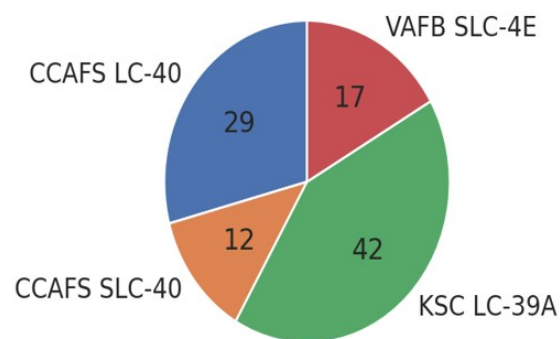
ALL SITES ▼

Payload range (kg)

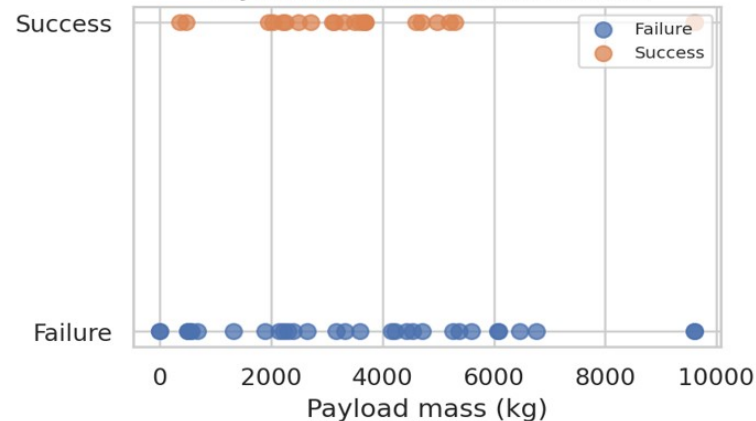
0

9600

Success count by site



Payload mass vs launch outcome

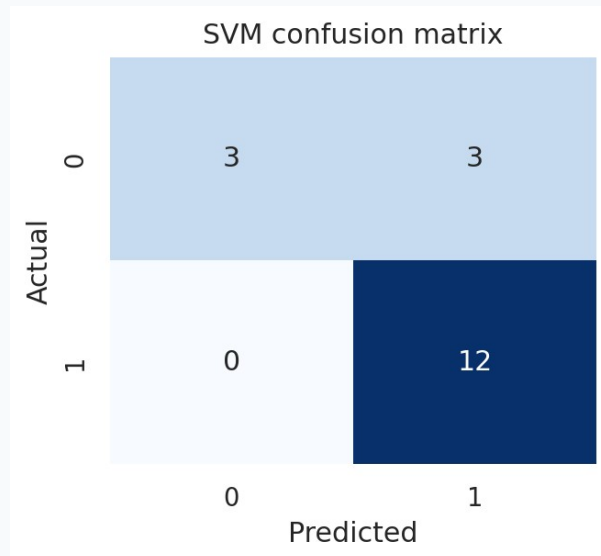
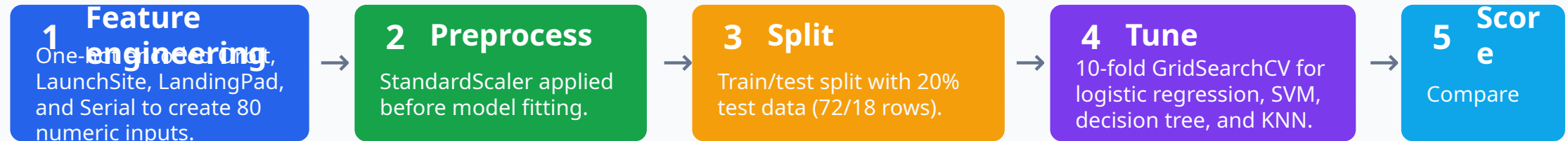


Interactive controls enable fast comparison of launch sites and payload windows.

What the dashboard adds: the dropdown isolates one site or compares all sites at once, and the payload slider turns the app into a quick "what range works best?" diagnostic.

Predictive analysis methodology

The classification stage converted the cleaned launch data into 80 numeric features and used model selection to compare multiple algorithms.

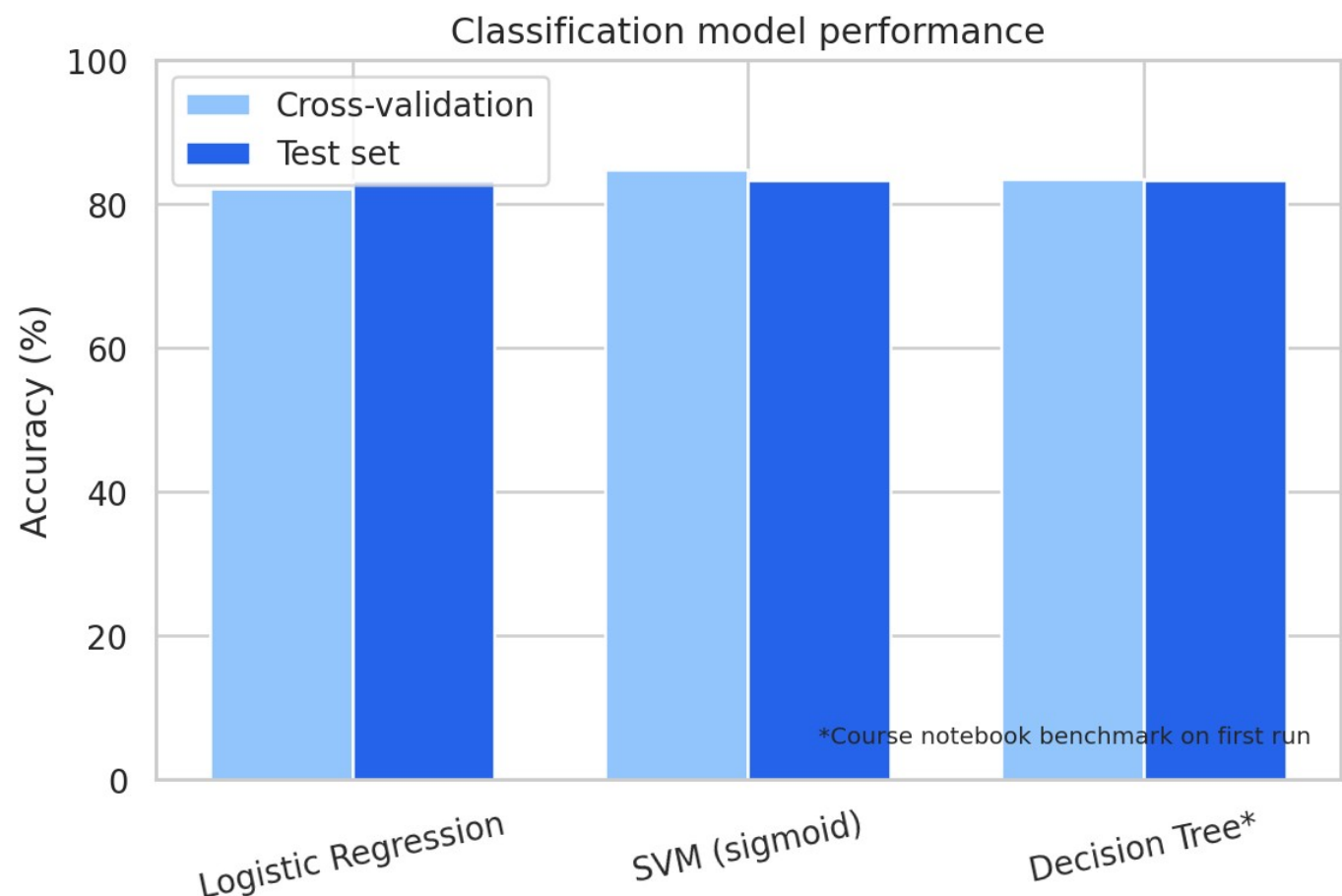


Modeling details worth calling out

- The course notebook's best SVM kernel on validation was sigmoid.
- The first course run reported 83.33% decision-tree test accuracy; reruns can shift because the tree search includes randomized settings.
- Using a held-out test set keeps the final score more honest than relying on cross-validation alone.

Predictive analysis results

The final comparison showed that the project's engineered features capture enough signal to support practical landing-success predictions.



**Best validation
model**
SVM with sigmoid
kernel

**Headline
benchmark**
85.53% course test
accuracy

Interpretati on

- A compact feature set built from site, orbit, payload, and booster context is enough to produce robust predictions on this dataset.
- The model is best understood as a decision-support tool: it highlights conditions associated with success rather than replacing mission engineering.

Conclusion and recommendations

The capstone tells a coherent operational story: landing success is learnable, explainable, and actionable when the right features are assembled.

What the project demonstrated

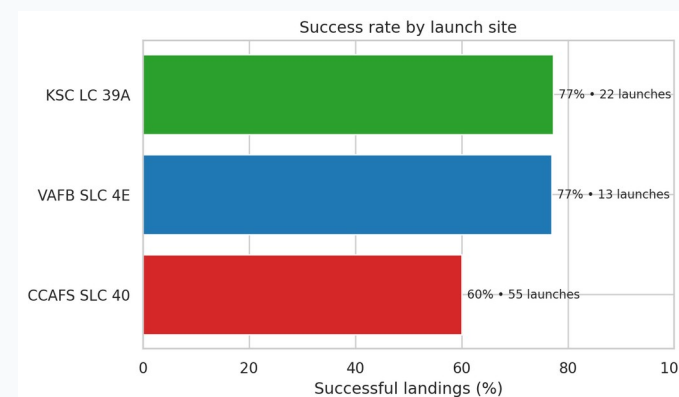
- Historical launch data can be cleaned into a clear supervised-learning problem without losing the operational meaning of the records.
- Static plots, SQL summaries, map views, and dashboards each surfaced different parts of the same story and reinforced one another.
- Machine learning made the story predictive by translating the cleaned signals into an 83%+ benchmark on the course test workflow.

Most important practical insight

Success is not driven by payload alone. Launch site, orbit, and mission context determine whether a payload profile is recoverable.

Suggested next steps

- Add newer launch data and retrain the classification pipeline on a refreshed sample.
- Test more explainability methods (feature importance, SHAP) to show why the model predicts success or failure for a given mission.
- Extend the dashboard with model predictions so stakeholders can interact with both descriptive and predictive views in one place.



Submission checklist and deliverables

This final slide makes the IBM rubric actionable before you export to PDF and submit.

Add your GitHub repository link before submission

Replace this placeholder with the repository that contains all completed notebooks and Python files:

<https://github.com/<your-username>/<your-spacex-capstone-repo>>

Rubric item	Covered in deck
Executive summary + introduction	Slides 2-3
Collection / wrangling methodology	Slides 4-5
EDA / SQL / map / dashboard results	Slides 6-10
Predictive analysis methodology + results	Slides 11-12
Conclusion + creativity / insights	Slide 13

Final export steps

- Review the deck once in PowerPoint or LibreOffice and verify that images, notes, and slide numbers are intact.
- Convert the presentation to PDF and upload the PDF, not the PPTX, if the rubric requires PDF for grading.
- Keep the notebooks / scripts in your GitHub repo so the submission has both the report and the code.

Ready-to-submit package from this workspace

- Final PowerPoint presentation
- Matching PDF export
- User-updated notebooks and generated visuals